OLIVIA SANTIAGO

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EDUCATION

Cornell University

Expected Graduation: May 2027

B.S. Mechanical Engineering
GPA: 3.703/4.30

COURSEWORK

- Dynamics
- Statics & Mechanics of Solids
- Thermodynamics
- Intro to Mechanical Design
- Spaceflight Mechanics'
- System Dynamics'
- Fluid Mechanics'
- Mechanics of Engineering Materials'

' in progress

SKILLS

Design

- SolidWorks
- Fusion 360
- MATLAB
- Python

Manufacturing

- Rapid Prototyping
- 3D Printing
- Laser Cutting
- Machining (Mill & Lathe)
- Composite layups

EXPERIENCE

AFRL Scholar | Air Force Research Laboratory Scholars Program at Edwards AFB

AFRL Scholars Program is a competitive research internship that develops the future STEM workforce by engaging students in hands-on defense R&D alongside AFRL engineers.

May 2025 - August 2025 | Edwards Air Force Base, CA

- Took ownership of a full-scope research project, independently leading reverse-engineering of adversary UAV performance using open-source intelligence (OSINT), enabling AFRL to benchmark foreign system capabilities against U.S. platforms
- Designed and programmed MATLAB-based dynamic trajectory simulation tools to model 3DOF flight dynamics and performance envelopes
- Integrated aerodynamic and propulsion data into simulations to evaluate climb performance and turn rate, producing results informing AFRL's threat assessment models
- Presented research findings to 20+ AFRL engineers, researchers, and industry partners, fielding technical questions and driving discussions on aerospace system performance
- Delivered actionable insights that were integrated into AFRL's adversary capability assessments, strengthening data-driving decision-making for future aerospace research priorities

Integration & Testing Sub Team Lead | Cornell University Unmanned Air Systems (CUAir)

CUAir is a student-run project team that designs, manufactures, tests, and programs a custom eVTOL unmanned aerial system.

Sept. 2024 - Present | Ithaca, NY

- Direct a 6-person subteam designing, manufacturing, and testing critical UAV subsystems, including the electrical bay, control surfaces, composite materials testing, wing loading, and tilt-rotor test stand, overseeing deliverables from concept through flight-ready implementation
- Engineered the UAV nose section's electrical bay for integration of dual 6s batteries on a 55 lb platform, reducing battery box weight by 3%, and accommodating complex geometry by designing custom 3D-printed components and laser-cut wood panels in CAD, collaborating closely with the electrical team; survived all test flights
- Converted an 1800 mm wingspan fixed-wing Skyhunter kit plane into a mechanically reinforced VTOL testing platform for a 35 lb UAV, designing and installing front carbon fiber booms, VTOL motor mounts, a side access panel, and modular wing joiners, along with custom add-ons to organize and improve accessibility of electrical components; validate autopilot and flight control algorithms

Machinist and Integration & Testing Engineer | Space Systems Design Studio (SSDS)

SSDS is a collaborative team of students designing and developing advanced spacecraft technologies, including CubeSats and chip satellite deployers, for flight experiments and space systems research.

Sept. 2024 - Present | Ithaca, NY

- Machined precise aluminum components for the DeSCENT ChipSat Deployer with end mills up to ⅛" in diameter to ∓.005" in tolerance, preparing the system to deploy 100 ChipSats on a Blue Origin suborbital mission.
- Conducted thermal testing of the Alpha CubeSat in a vacuum chamber while monitoring component temperatures with sensors, and performed light sail deployment and structural tests to validate flight hardware for mission success
- Supported cleanroom assembly and final integration of Alpha CubeSat hardware for launch aboard SpaceX NG-23 resupply mission to the ISS

Machine Shop Trainer | Manufacturing Learning Studio (MLS)

Cornell Engineering's machine shop that supports coursework, research, and project teams.

Aug. 2024 - Present | Ithaca, NY

- Train students twice a week on safe and effective use of manual mills and lathes, including operations such as facing, side milling, drilling, chamfering, tapping, and threading with a die.